Alternating multilinear maps and wedge product

Soham Saha

https://soham-saha.github.io

https://soham-saha.codeberg.page

We will be working over $\mathbf{Vect}_k$ for some arbitrary field k of characteristic 0.

We will be needing to make two choices while we set everything up. They have been marked in the notes as **1st choice** and **2nd choice**.

Using the machinery developed in the previous section about groups acting on vector spaces, we have that:

- Consider the action of S_n on $T^n(V)$ by

$$\sigma(v_1 \otimes ... \otimes v_n) = \text{sgn}(\sigma)\left(v_{\sigma^{-1}(1)} \otimes ... \otimes v_{\sigma^{-1}(n)}\right).$$

  Then, linear maps from $T^n(V)$ to W (i.e., multilinear maps from $(V, ..., V)$ to W) are called "alternating" if they respect this action; i.e.,

$$\varphi \in \text{Hom}_k(T^n(V), W) \text{ is called alternating iff}$$
$$\forall v \in T^n(V), \forall \sigma \in S_n, \varphi(\sigma(v)) = \varphi(v).$$

- Define the nth wedge product of V as

$$\Lambda^n(V) := \frac{T^n(V)}{\text{span}_k(\{v - w : v \in T^n(V), w \in \text{orbit}(v)\})}$$

  $\Lambda^n(V)$ has the nice property that alternating multilinear maps from $(V, ..., V)$ to W are in one-to-one correspondence with linear maps from $\Lambda^n(V)$ to W. So, we have

$$\text{AltMulti}(V, ..., V; W) \cong \text{Hom}_k(\Lambda^n(V), W)$$

- We write $v_1 \wedge ... \wedge v_n \in \Lambda^n(V)$ for the image of $v_1 \otimes ... \otimes v_n \in T^n(V)$ under the quotient map.
- A vector in $T^n(V)$ is called an "alternating tensor" if it is stable under the action of S_n; i.e., it has a singleton orbit; i.e.,

$$\forall \sigma \in S_n, \sigma(v) = v.$$

  We shall denote the set of alternating tensors in $T^n(V)$ as V^n_{alt}.

- Consider the map

$$\text{Alt}^n \in \text{End}_k(T^n(V))$$
$$\forall v \in T^n(V), \text{Alt}^n(v) := \frac{1}{n!} \sum_{\sigma \in S_n} \sigma(v).$$

  Note that in the above definition, $\sigma(v) = \sigma(v_1 \otimes ... \otimes v_n) = \text{sgn}(\sigma) \times v_{\sigma^{-1}(1)} \otimes ... \otimes v_{\sigma^{-1}(n)}$, since that is how σ acts on v.

  It is easy to show that $\text{im}(\text{Alt}^n) = V^n_{\text{alt}}$.

This gives us a characterization of how the alternating tensors *look* like.

Our definition of Alt^n boils down to

$$v_1 \otimes ... \otimes v_n \mapsto \frac{1}{n!} \sum_{\sigma \in S_n} \text{sgn}(\sigma) \Big(v_{\sigma^{-1}(1)} \otimes ... \otimes v_{\sigma^{-1}(n)} \Big)$$

Note that some authors define Alt^n as

$$v_1 \otimes ... \otimes v_n \mapsto \frac{1}{n!} \sum_{\sigma \in S_n} \text{sgn}(\sigma) \Big(v_{\sigma(1)} \otimes ... \otimes v_{\sigma(n)} \Big),$$

but don't worry, since we are taking sum over the entirety of S_n, so σ and σ^{-1} come in pairs; and $\text{sgn}(\sigma) = \text{sgn}(\sigma^{-1})$. Both definitions are equivalent.

- Alt^n is a projection operator on $T^n(V)$ (projecting onto V^n_{alt}), and we have

  $$\ker(\text{Alt}^n) = \text{span}_k(\{v - w : v \in T^n(V), w \in \text{orbit}(v)\}).$$

  Therefore,

  $$\Lambda^n(V) = \frac{T^n(V)}{\ker(\text{Alt}^n)} \cong \text{im}(\text{Alt}^n) = V^n_{\text{alt}},$$

  using the first isomorphism theorem.

Explicitly writing the isomorphism out, we have

$$\left([v] \mapsto \text{Alt}^n(v) = \frac{1}{n!} \sum_{\sigma \in S_n} \sigma(v) \right) : \Lambda^n(V) \to V^n_{\text{alt}}$$
$$(v \mapsto [v]) : V^n_{\text{alt}} \to \Lambda^n(V)$$

Again, note that $\sigma(v) = \sigma(v_1 \otimes ... \otimes v_n) = \text{sgn}(\sigma) \times v_{\sigma^{-1}(1)} \otimes ... \otimes v_{\sigma^{-1}(n)}$, since that is how σ acts on v.

I believe that Spivak does **not** use this isomorphism, and instead, chooses

$$[v] \mapsto n! \text{Alt}^n(v)$$
$$v \mapsto \frac{1}{n!}[v]$$

Details about why this might be so are described below when we try to define the product on $\bigoplus_{n \geq 0} V^n_{\text{alt}}$.

This is the **1st choice** we have to make. Choosing the isomorphism $[v] \mapsto \text{Alt}^n(v)$ leads to the wedge product becoming

$$v \in V^k_{\text{alt}}, w \in V^l_{\text{alt}}$$
$$v \wedge w := \text{Alt}^{k+l}(v \otimes w) \in V^{k+l}_{\text{alt}},$$

while choosing the isomorphism $[v] \mapsto n! \text{Alt}^n(v)$ leads to the wedge product becoming

$$v \in V_{\text{alt}}^k, w \in V_{\text{alt}}^l$$
$$v \wedge w := \binom{k+l}{l} \text{Alt}^{k+l}(v \otimes w) \in V_{\text{alt}}^{k+l}$$

Note that the **1st choice** does *not* affect the multiplication on $\bigoplus_{n\geq 0} \Lambda^n(V)$, it only affects the multiplication on $\bigoplus_{n\geq 0} V_{\text{alt}}^n$.

I shall leave this choice undetermined for now, since I am not very sure which side I want to commit to. If I ever need to use either the isomorphism between $\Lambda^n(V)$ and V_{alt}^n or the product on $\bigoplus_{n\geq 0} V_{\text{alt}}^n$, I will update this part to decide which choice I shall use.

Suppose V is a finite dimensional vector space with $\dim(V) = d$.

If $n > d$, then the subspace we are quotienting by in the expression for $\Lambda^n(V)$ is the entire space $T^n(V)$, and therefore $\dim(\Lambda^n V) = 0$.

If $n \leq d$, and V has a basis $\{v_1, ..., v_d\}$, then $\left\{v_{i_1} \wedge ... \wedge v_{i_n} : 1 \leq i_1 < i_2 < ... < i_n \leq d\right\}$ is a basis for $\Lambda^n(V)$. So,

$$n \leq d \Rightarrow \dim(\Lambda^n(V)) = \binom{\dim(V)}{n}.$$

Particularly, $\Lambda^{\dim(V)}(V)$ is isomorphic to the scalar field k.

Often we are interested in multilinear maps from $(V, ..., V)$ back to the base field k. This corresponds to

$$\text{Hom}_k(\Lambda^n(V), k) = (\Lambda^n(V))^*$$

Recall that, if V is a finite dimensional vector space, then we have

$$T^n(V^*) \cong (T^n(V))^*$$

The quotient map $\pi_1 : T^n(V) \twoheadrightarrow \Lambda^n(V)$ is a surjection, which induces an inclusion $\pi_1^* : \Lambda^n(V)^* \hookrightarrow T^n(V)^*$, and $(T^n(V))^* \cong T^n(V^*)$, which gives us the inclusion

$$\Lambda^n(V)^* \hookrightarrow T^n(V^*)$$

Again, assume V is finite dimensional.

Observe the following chain of morphisms:

$$\Lambda^n(V)^* \overset{\pi_1^*}{\hookrightarrow} T^n(V)^* \overset{i}{\to} T^n(V^*) \overset{\pi_2}{\twoheadrightarrow} \Lambda^n(V^*),$$

where $i : T^n(V)^* \xrightarrow{\sim} T^n(V^*)$ is from the canonical isomorphism $T^n(V)^* \cong T^n(V^*)$ (described above in **Tensor product miscellany**).

This chain of morphisms composes to the following map from $\Lambda^n(V)^*$ to $\Lambda^n(V^*)$:

$$f \mapsto \sum_{1\leq i_1<...<i_n\leq d} n! f\left(e_{i_1} \wedge ... \wedge e_{i_n}\right) \cdot \left(e^{i_1} \wedge ... \wedge e^{i_n}\right) = \sum_{|I|=n} n! f(e_I) e^I,$$

where $\{e_i\}_{1\leq i\leq d}$ is a basis for V and $\{e^i\}_{1\leq i\leq d}$ is the dual basis.

It turns out that the above is indeed an isomorphism, and the inverse of this isomorphism is

$$\left[f_1 \wedge ... \wedge f_n \mapsto \left[u_1 \wedge ... \wedge u_n \mapsto \frac{1}{n!} \det\left[f_i(u_j)\right]\right]\right] : \Lambda^n(V^*) \xrightarrow{\sim} \Lambda^n(V)^*$$

Using this pair of isomorphisms is known as the **Alt convention**.

Instead of keeping around the annoying $\frac{1}{n!}\det[-]$, we can instead choose a different isomorphism between $\Lambda^n(V)^*$ and $\Lambda^n(V^*)$

$$f \mapsto \sum_{|I|=n} f(e_I)e^I$$

$$f_1 \wedge ... \wedge f_n \mapsto \left[u_1 \wedge ... \wedge u_n \mapsto \det\left[f_i(u_j)\right]\right]$$

This is known as the **determinant convention**.

The difference between the "Alt convention" and the "determinant convention" is mentioned in *Introduction to Riemannian Manifolds* (2nd edition) [John M. Lee ("Differential forms and integration" section of Appendix B)], and the scaling by $n!$ issue is mentioned in section 44.5 of *An Infinitely Large Napkin* (Evan Chen, version: v1.6.20250213).

This is the **2nd choice** we have to make, and it will determine what isomorphism we shall use between $\Lambda^n(V^*)$ and $\Lambda^n(V)^*$

Moving onwards, we will be using the **determinant convention**.

The Leibniz determinant formula states that:

$$\det\left(\left[a_{ij}\right]_{n\times n}\right) = \sum_{\sigma\in S_n} \operatorname{sgn}(\sigma) a_{1\sigma(1)}...a_{n\sigma(n)}$$

Now, we would like to describe multiplication over $\bigoplus_{n\geq 0}\Lambda^n(V)$ and over $\bigoplus_{n\geq 0} V^n_{\text{alt}}$.

For both multiplications, we would be using the same symbol "$\wedge$".

Recall that $\Lambda^n(V)$ is defined to be a quotient of $T^n(V)$, and we have already defined multiplication over $\oplus_{n\geq 0} T^n(V)$, so for $[v_1] \in \Lambda^k(V), [v_2] \in \Lambda^l(V)$ (for some $v_1 \in T^k(V), v_2 \in T^l(V)$), we define the product of $[v_1]$ and $[v_2]$ as

$$[v_1] \wedge [v_2] = [v_1 \otimes v_2] \in \Lambda^{k+l}(V)$$

Now, we can use the isomorphism $\Lambda^n(V) \cong V^n_{\text{alt}}$ to get a multiplication on $\bigoplus_{n\geq 0} V^n_{\text{alt}}$.

Recall that the isomorphism that we gained canonically from the fact that the Reynolds operator (Alt^n) is a projection was

$$[v] \in \Lambda^n(V) \mapsto \mathrm{Alt}^n(v) \in V_{\mathrm{alt}}^n$$
$$v \in V_{\mathrm{alt}}^n \mapsto [v] \in \Lambda^n(V)$$

Let us try using this isomorphism to get multiplication on $\bigoplus_{n\geq 0} V_{\mathrm{alt}}^n$:

Given $v \in V_{\mathrm{alt}}^k, w \in V_{\mathrm{alt}}^l$, they correspond to $[v] \in \Lambda^k(V), [w] \in \Lambda^l(V)$, which multiply to give $[v \otimes w] \in \Lambda^{k+l}(V)$, which corresponds to $\mathrm{Alt}^{k+l}(v \otimes w) = \frac{1}{(k+l)!}\sum_{\sigma \in S_{k+l}} \sigma(v \otimes w)$. So, the multiplication we get is

$$v \wedge w = \mathrm{Alt}^{k+l}(v \otimes w)$$

The above formulation doesn't have any $\binom{k+l}{k}$ scalar in it, while Spivak does. My guess at how that might be arising is that instead of using the above isomorphism, Spivak uses $v \mapsto \frac{1}{n!}[v]; [v] \mapsto n!\mathrm{Alt}^n(v)$. If we use *that* isomorphism, then

Given $v \in V_{\mathrm{alt}}^k, w \in V_{\mathrm{alt}}^l$, they correspond to $\frac{1}{k!}[v] \in \Lambda^k(V), \frac{1}{l!}[w] \in \Lambda^l(V)$, which multiply to give $\frac{1}{k!l!}[v \otimes w] \in \Lambda^{k+l}(V)$, which corresponds to $\binom{k+l}{k}\mathrm{Alt}^{k+l}(v \otimes w) \in V_{\mathrm{alt}}^{k+l}$. So, the multiplication we get is

$$v \wedge w = \binom{k+l}{k}\mathrm{Alt}^{k+l}(v \otimes w)$$

Have a look at the description of **1st choice** above.

Given basis $\{e_1, ..., e_d\}$ of V, and vectors $v_1, ..., v_d \in V$ such that

$$[e_1 \ \ldots \ e_d]A_{d\times d} = [v_1 \ \ldots \ v_d],$$

we have $v_1 \wedge ... \wedge v_d \in \Lambda^d(V)$, and simplifying, we get that

$$v_1 \wedge ... \wedge v_d = \det(A) \cdot e_1 \wedge ... \wedge e_d$$

Given a k-linear map $\varphi : V \to W$, we have a map $T^n(\varphi) : T^n(V) \to T^n(W)$ given by $(v_1 \otimes ... \otimes v_n) \mapsto (\varphi(v_1) \otimes ... \otimes \varphi(v_n))$.

Now, $\pi \circ T^n(\varphi) := (v_1 \otimes ... \otimes v_n) \mapsto \varphi(v_1) \wedge ... \wedge \varphi(v_n)$ is an alternating map from $T^n(V)$ to $\Lambda^n(W)$, and so we get an unique map $\Lambda^n(\varphi) : \Lambda^n(V) \to \Lambda^n(W)$.

This shows that Λ^n is functorial, just like T^n.

Note that Λ^0 is also functorial, with the object map being $A \mapsto (\Lambda^0 A = k)$ and $f \in \mathrm{Hom}(A, B) \mapsto (\Lambda^0 f = \mathrm{id}_k) \in \mathrm{Hom}(k, k)$.

Given basis $B_V = \{u_1, ..., u_k\}$ for V and $B_W = \{w_1, ..., w_l\}$ for W. Suppose the matrix of φ with respect to this basis is $A_{l\times k}$.

Suppose $n \leq \min(k, l)$. Then we have basis $B_V^0 = \left\{u_{i_1} \wedge ... \wedge u_{i_n} : 1 \leq i_1 < ... < i_n \leq k\right\}$ for $\Lambda^n(V)$ and $B_W^0 = \left\{w_{j_1} \wedge ... \wedge w_{j_n} : 1 \leq j_1 < ... < j_n \leq l\right\}$ for $\Lambda^n(W)$. Then, for some fixed $1 \leq i_1 < ... < i_n \leq k$, we have

$$\Lambda^n(\varphi)\left(u_{i_1} \wedge ... \wedge u_{i_n}\right) = \varphi\left(u_{i_1}\right) \wedge ... \wedge \varphi\left(u_{i_n}\right)$$

Fix some $1 \leq j_1 < ... < j_n \leq l$. We can write $\Lambda^n(\varphi)\left(u_{i_1} \wedge ... \wedge u_{i_n}\right)$ in the basis B_W^0, and the coordinate along $w_{j_1} \wedge ... \wedge w_{j_n}$ will be

$$\begin{aligned}
&\sum_{\sigma \in S_n} \left(\prod w^{j_{\sigma(k)}}\left(\varphi\left(u_{i_k}\right)\right)\right) \cdot w_{j_{\sigma(1)}} \wedge ... \wedge w_{j_{\sigma(n)}} \\
&= \sum_{\sigma \in S_n} \operatorname{sgn}(\sigma) \cdot \left(\prod A_{\sigma(k),k}\right) \cdot w_{j_1} \wedge ... \wedge w_{j_n} \\
&= \det(\text{minor}_{n \times n}(A, (i_1, ..., i_n), (j_1, ..., j_n))) \cdot w_{j_1} \wedge ... \wedge w_{j_n}
\end{aligned}$$

So, the matrix of $\Lambda^n(\varphi)$ will have determinants of $n \times n$ minors of A as its entries.

Therefore, given $\varphi : V \to V$ with matrix A wrt basis $\{e_1, ..., e_d\}$, $\Lambda^d\varphi$ will have matrix $[\det(A)]$ with respect to the basis $\{e_1 \wedge ... \wedge e_d\}$.

But now, a nice thing happens. $\Lambda^d(V)$ is 1-dimensional, and therefore the only data that $\Lambda^d(\varphi)$ stores is a single scalar value, which in this case happens to be $\det(A)$.

So, given $\varphi \in \text{End}_k(V)$, $\Lambda^d(\varphi) \in \text{End}_k\left(\Lambda^d(V)\right)$ is the map "scaling by $\det(\varphi)$".

This gives us a basis free definition of $\det(\varphi)$, so we can take **determinants of endomorphisms over finite dimensional vector spaces.**

The following discussion is when the base field k is equal to $\mathbb{R}$.

When do two basis $B_u = \{u_1, ..., u_d\}$ and $B_v = \{v_1, ..., v_d\}$ of a vector space V have the **same orientation**?

1st try When the change of basis matrix from B_u to B_v has positive determinant.

2nd try $u_1 \wedge ... \wedge u_d$ and $v_1 \wedge ... \wedge v_d$ are both vectors in the 1-dimensional vector space $\Lambda^d(V)$, so each of them is a scalar multiple of the other. If the scalar is positive, then they have same orientation; otherwise not.

3rd try For every $\omega \in \Lambda^d(V)^*$, we have $\omega(u_1 \wedge ... \wedge u_d) = \det(A)\omega(v_1 \wedge ... \wedge v_d)$, if $[v_1 \ \ ... \ \ v_d]A = [u_1 \ \ ... \ \ u_d]$. Define that B_u and B_v have same orientation iff

$$\forall \omega \in \Lambda^d(V)^*, \omega(u_1 \wedge ... \wedge u_d) \text{ and } \omega(v_1 \wedge ... \wedge v_d) \text{ have the same sign.}$$
